

EXAMPLE 2. The angle of elevation of a tower at a point is 45° . After going 40 m towards the foot of the tower, the angle of elevation of the tower becomes 60° . Then, the height of the tower is

- a. $20(\sqrt{3} + 1)$ m b. $\frac{20\sqrt{3}}{\sqrt{3} + 1}$ m
c. $\frac{40\sqrt{3}}{\sqrt{3} + 1}$ m d. $\frac{40\sqrt{3}}{\sqrt{3} - 1}$ m

Sol. d. Let $PQ (= h)$ be the height of the tower. Let S and R be the points where the angles subtended are 45° and 60° , respectively.

In right angled ΔPQR ,

$$\tan 60^\circ = \frac{PQ}{RQ}$$

$$\Rightarrow \sqrt{3} = \frac{h}{x} \Rightarrow x = \frac{h}{\sqrt{3}} \quad \dots(i)$$

In right angled ΔPQS , $\tan 45^\circ = \frac{PQ}{SQ} = \frac{PQ}{SR + RQ} = \frac{h}{40 + x}$

$$\Rightarrow h = 40 + x \Rightarrow h = 40 + \frac{h}{\sqrt{3}} \quad [\text{from Eq. (i)}]$$

$$\Rightarrow h \left(1 - \frac{1}{\sqrt{3}}\right) = 40 \Rightarrow h \left(\frac{\sqrt{3} - 1}{\sqrt{3}}\right) = 40$$

$$\therefore h = \frac{40\sqrt{3}}{\sqrt{3} - 1} \text{ m}$$

EXAMPLE 3. The angles of elevation of the top of a tower from two points at distances ' a ' and ' b ' ($a > b$) from the base and in the same straight line with it are complementary. Then, the height of the tower is

- a. $\frac{1}{\sqrt{ab}}$ b. $\sqrt{ab}$ c. $\frac{a}{b}$ d. ab

Sol. b. Let $AD = h$ be the height of the tower and the angle of elevation of the top of tower at B and C be $(90^\circ - \theta)$ and θ , respectively.

In right angled ΔADC ,

$$\tan \theta = \frac{AD}{CD} = \frac{h}{b} \quad \left[\because \tan \theta = \frac{P}{B} \right]$$

$$h = b \tan \theta \quad \dots(i)$$

In right angled ΔADB , $\tan (90^\circ - \theta) = \frac{AD}{BD} \Rightarrow \cot \theta = \frac{h}{a}$
[$\because \tan (90^\circ - \theta) = \cot \theta$]

$$\Rightarrow \frac{h}{a} = \cot \theta \Rightarrow \frac{a}{h} = \tan \theta \quad \dots(ii)$$

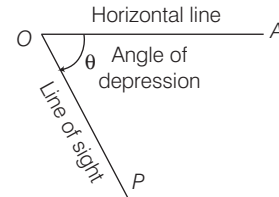
From Eqs. (i) and (ii), we get

$$h = b \left(\frac{a}{h}\right) \Rightarrow h^2 = ab$$

$$\therefore h = \sqrt{ab}$$

ANGLE OF DEPRESSION

Angle of depression is defined as an angle formed by the line of sight with the horizontal line when an object is viewed in the downward direction.



Let P be the position of the object below the horizontal line OA and O be the eye of the observer, then $\angle AOP$ is called angle of depression.

IMPORTANT POINTS

- The angle of depression of a point P as seen from a point O is numerically equal to the angle of elevation of O as seen from P .
- Angle of elevation and angle of depression are always acute angle.
- Unless mentioned, the height of the observer is not considered.

EXAMPLE 4. The angle of depression of two ships from the top of a light house are 45° and 30° towards East. If the ships are 200 m apart, the height of the light house is

- a. $\frac{200}{\sqrt{3} - 1}$ m b. $\frac{200}{\sqrt{3} + 1}$ m c. $\frac{100}{\sqrt{3} - 1}$ m d. $\frac{100}{\sqrt{3} + 1}$ m

Sol. a. Let $AB (= h)$ be the height of the light house and D, C be the ships.

Given, $DC = 200$ m

$$\angle ADC = \angle XAD = 30^\circ$$

$$\text{and } \angle ACB = \angle XAC = 45^\circ$$

Now, in ΔACB ,

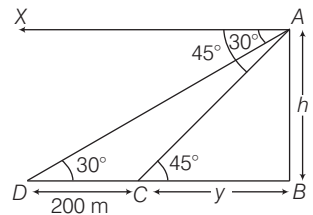
$$\tan 45^\circ = \frac{AB}{CB} \Rightarrow 1 = \frac{h}{y} \Rightarrow h = y \quad \dots(i)$$

$$\text{and in } \Delta ADB, \tan 30^\circ = \frac{AB}{BD} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{200 + y}$$

$$\Rightarrow 200 + y = \sqrt{3} h \Rightarrow 200 + h = \sqrt{3} h \quad [\text{from Eq. (i)}]$$

$$\Rightarrow 200 = \sqrt{3} h - h \Rightarrow 200 = h(\sqrt{3} - 1)$$

$$\therefore h = \left(\frac{200}{\sqrt{3} - 1}\right) \text{ m}$$



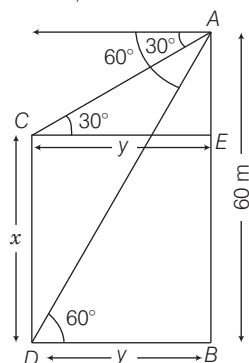
EXAMPLE 5. From the top of a building 60 m high the angles of depression of the top and bottom of a tower are observed to be 30° and 60° . Then, the height of the tower is

- a. 10 m b. 20 m c. 30 m d. 40 m

Sol. d. Let x be the height of the tower.

Here, $AB = 60$ m and $BD = y$ m

In right angled $\triangle ABD$,



$$\frac{AB}{BD} = \tan 60^\circ \Rightarrow \frac{60}{y} = \sqrt{3}$$

$$\Rightarrow y = \frac{60}{\sqrt{3}} \quad \dots(i)$$

In right angled $\triangle AEC$,

$$\frac{AE}{EC} = \tan 30^\circ \Rightarrow \frac{AE}{BD} = \frac{1}{\sqrt{3}} \quad [\because CE = DB]$$

$$\Rightarrow \frac{AE}{y} = \frac{1}{\sqrt{3}} \Rightarrow \frac{AE}{60} = \frac{1}{\sqrt{3}} \quad [\text{from Eq. (i)}]$$

$$\Rightarrow AE = \frac{1}{\sqrt{3}} \times \frac{60}{\sqrt{3}} = 20 \text{ m} \quad [\because \sqrt{3} \times \sqrt{3} = 3]$$

$$\therefore x = AB - AE = 60 - 20 = 40 \text{ m}$$

Hence, the height of tower is 40 m

> PRACTICE EXERCISE

- The length of shadow of a tree is 16 m when the angle of elevation of the Sun is 60° . What is the height of the tree?
(a) 8 m (b) 16 m (c) $16\sqrt{3}$ m (d) $\sqrt{3}$ m
- From a light house the angles of depression of two ships on opposite sides of the light house are observed to 30° and 45° . If the height of light house is h , then what is the distance between the ships?
(a) $(\sqrt{3} + 1)h$ (b) $(\sqrt{3} - 1)h$ (c) $\sqrt{3}h$ (d) $\left(1 + \frac{1}{\sqrt{3}}\right)h$
- The angle of elevation of the top of a tower from the bottom of a building is twice that from its top. What is the height of the building, if the height of the tower is 75 m and the angle of elevation of the top of the tower from the bottom of the building is 60° ?
(a) 25 m (b) 37.5 m (c) 50 m (d) 60 m
- A radio transmitter antenna of height 100 m stands at the top of a tall building. At a point on the ground, the angle of elevation of bottom of the antenna is 45° and that of top of antenna is 60° . What is the height of the building?
(a) 100 m (b) 50 m (c) $50(\sqrt{3} + 1)$ m (d) $25\sqrt{3}$ m
- The shadow of a tower is 15 m when the Sun's altitude is 30° . What is the length of the shadow when the Sun's altitude is 60° ?
(a) 3 m (b) 4 m (c) 5 m (d) 6 m
- The angle of elevation of a moon when the length of the shadow of a pole is equal to its height is
(a) 60° (b) 45° (c) 90° (d) 30°
- An aeroplane flying horizontally 1 km above the ground is observed at an elevation of 60° . After 10 s, its elevation is observed to be 30° , the uniform speed of the aeroplane in km/h is
(a) $\frac{240}{\sqrt{2}}$ (b) $240\sqrt{3}$
(c) 240 (d) $260\sqrt{3}$
- A person of height 2m wants to get a fruit which is on a pole of height $(10/3)$ m. If he stands at a distance of $(4/\sqrt{3})$ m from the foot of the pole, then the angle at which he should throw the stone, so that it hits the fruit is
(a) 60° (b) 45° (c) 90° (d) 30°
- A vertical tower stands on a horizontal plane and is surmounted by a vertical flagstaff of height h . At a point on the plane, the angle of elevation of the bottom of the flagstaff is α and that of the top of the flagstaff is β . Then, the height of the tower is
(a) $\frac{h \tan \beta}{\tan \alpha - \tan \beta}$ (b) $\frac{h \tan \alpha}{\tan \alpha + \tan \beta}$
(c) $\frac{h \tan \beta}{\tan \alpha + \tan \beta}$ (d) $\frac{h \tan \alpha}{\tan \beta - \tan \alpha}$
- From a window (h m high above the ground) of a house in a street, the angle of elevation and depression of the top and the foot of another house on the opposite side of the street are θ and ϕ , respectively. Then, the height of the opposite house is
(a) $h \tan \theta \cot \phi$ (b) $h [\tan \theta \cot \phi + 1]$
(c) $h [\cot \theta \tan \phi + 1]$ (d) $h \cot \theta \tan \phi$

- 11.** The length of the shadow of a person s cm tall when the angle of elevation of the Sun is α is p cm. It is q cm when the angle of elevation of the Sun is β . Which one of the following is correct when $\beta = 3\alpha$?
- (a) $p - q = s \left(\frac{\tan \alpha - \tan 3\alpha}{\tan 3\alpha \tan \alpha} \right)$ (b) $p - q = s \left(\frac{\tan 3\alpha - \tan \alpha}{3 \tan 3\alpha \tan \alpha} \right)$
 (c) $p - q = s \left(\frac{\tan 3\alpha - \tan \alpha}{\tan 3\alpha \tan \alpha} \right)$ (d) $p - q = s \left(\frac{\tan 2\alpha}{\tan 3\alpha \tan \alpha} \right)$
- 12.** Two posts are ' K ' m apart and the height of one is double that of the other. If from the middle point of the line joining their feet, an observer finds the angular elevations of their tops to be complementary, then the height (in metre) of the shorter post is
- (a) $\sqrt{3} K$ (b) $\frac{K}{2\sqrt{2}}$ (c) $2 K$ (d) $\frac{K}{\sqrt{2}}$
- 13.** A flagstaff 5 m high stands on a building of 25 m height. At an observer at a height of 30 m, the flagstaff and the building subtend equal angles. The distance of the observer from the top of the flagstaff is
- (a) $\frac{3}{2}$ m (b) $3\sqrt{\frac{5}{2}}$ m (c) $\frac{5\sqrt{3}}{2}$ m (d) $5\sqrt{\frac{3}{2}}$ m
- 14.** A man is watching from the top of a tower a boat speeding away from the tower. The boat makes an angle of depression of 45° with the man's eye when at a distance of 60 m from the bottom of tower. After 5 s, the angle of depression becomes 30° . What is the approximate speed of the boat assuming that it is running in still water?
- (a) 31.5 km/h (b) 36.5 km/h (c) 38.5 km/h (d) 40.5 km/h
- 15.** Suppose the angle of elevation of the top of a tree at a point E due East of the tree is 60° and that at a point F due West of the tree is 30° . If the distance between the points E and F is 160 ft, then what is the height of the tree?
- (a) $40\sqrt{3}$ ft (b) 60 ft (c) $\frac{40}{\sqrt{3}}$ ft (d) 23 ft
- 16.** A telegraph post gets broken at a point against a storm and its top touches the ground at a distance 20 m from the base of the post making an angle 30° with the ground. What is the height of the post?
- (a) $\frac{40}{\sqrt{3}}$ m (b) $20\sqrt{3}$ m (c) $40\sqrt{3}$ m (d) 30 m
- 17.** If the angle of elevation of a cloud from a point h m above a lake is β and the angle of depression of its reflection in the lake is α , then the height of the cloud is
- (a) $\frac{h \operatorname{cosec}(\alpha - \beta)}{\operatorname{cosec}(\alpha - \beta)}$ (b) $h \operatorname{cosec}(\alpha - \beta) \sin(\alpha - \beta)$
 (c) $h \sin(\alpha + \beta) \operatorname{cosec}(\alpha - \beta)$ (d) $\frac{h \operatorname{cosec}(\alpha + \beta)}{\sin(\alpha - \beta)}$
- 18.** The angle of elevation of the top of an incomplete vertical pillar at a horizontal distance of 100 m from its base is 45° . If the angle of elevation of the top of complete pillar at the same point is to be 60° , then the height of the incomplete pillar is to be increased by
- (a) $50\sqrt{2}$ m (b) 100 m
 (c) $100(\sqrt{3} - 1)$ m (d) $100(\sqrt{3} + 1)$ m
- 19.** A man on the top of a vertical observation tower observes a car moving at a uniform speed coming directly towards it. If it takes 12 min for the angle of depression to change from 30° to 45° , how soon after this will the car reach the observation tower?
- (a) 14 min 35 s (b) 15 min 49 s
 (c) 16 min 23 s (d) 18 min 5 s
- 20.** You are stationed at a radar base and you observe an unidentified plane at an altitude $h = 1000$ m flying towards your radar base at an angle of elevation $= 30^\circ$. After exactly one minute, your radar sweep reveals that the plane is now at an angle of elevation $= 60^\circ$ maintaining the same altitude. What is the speed (in m/s) of the plane?
- (a) 15.58 m/s (b) 19.25 m/s
 (c) 18 m/s (d) 11.25 m/s
- 21.** The angle of elevation of the top of the tower from a point on the ground is $\sin^{-1}(3/5)$. If the point of observation is 20 m away from the foot of the tower, what is the height of the tower?
- (a) 9 m (b) 18 m (c) 15 m (d) 12 m
- 22.** A balloon of radius r makes an angle α at the eye of an observer and the angle of elevation of its centre is β . The height of its centre from the ground level is given by
- (a) $r \sin \beta \operatorname{cosec} \alpha / 2$ (b) $r \operatorname{cosec} \alpha / 2 \sin \alpha$
 (c) $r \operatorname{cosec} \alpha \sin \beta$ (d) None of these
- Directions** (Q. Nos. 23-25) The height of a tower is h and the angle of elevation of the top of the tower is α . When observer on moving a distance $h/2$ towards the tower, the angle of elevation becomes β .
- 23.** What is the value of $\cot \alpha - \cot \beta$?
- (a) $1/2$ (b) $2/3$ (c) 1 (d) 2
- 24.** If $\alpha = 30^\circ$, then find the value of $\cot \beta$.
- (a) $\sqrt{3} - 1/2$ (b) $\frac{2}{2\sqrt{3} - 2}$
 (c) $\frac{1}{1 + \sqrt{3}}$ (d) $\frac{1}{1 - \sqrt{3}}$
- 25.** If $\tan \beta = 4$, then distance between observer and tower when angle of elevation is β
- (a) h (b) $h/4$ (c) $\frac{3h}{4}$ (d) $\frac{h}{3}$

PREVIOUS YEARS' QUESTIONS

- 26.** What is the angle of elevation of the Sun when the shadow of a pole is $\sqrt{3}$ times the length of the pole? ☑ 2012 I
 (a) 30° (b) 45°
 (c) 60° (d) None of these
- 27.** From the top of a cliff 200 m high, the angles of depression of the top and bottom of a tower are observed to be 30° and 45° , respectively. What is the height of the tower? ☑ 2012 I
 (a) 400 m (b) $400\sqrt{3}$ m
 (c) $400/\sqrt{3}$ m (d) None of these
- 28.** The angles of elevation of the top of a tower from two points which are at distances of 10 m and 5 m from the base of the tower and in the same straight line with it are complementary. The height of the tower is ☑ 2012 II
 (a) 5 m (b) 15 m (c) $\sqrt{50}$ m (d) $\sqrt{75}$ m
- 29.** A ladder 20 m long is placed against a wall, so that the foot of the ladder is 10 m from the wall. The angle of inclination of the ladder to the horizontal will be ☑ 2012 II
 (a) 30° (b) 45°
 (c) 60° (d) 75°

Directions (Q. Nos. 30-33) Read the following information carefully to answer the questions that follow.

As seen from the top and bottom of a building of height h m, the angles of elevation of the top of a tower of

height $\frac{(3 + \sqrt{3})h}{2}$ m are α and β , respectively. ☑ 2013 I

- 30.** If $\beta = 30^\circ$, then what is the value of $\tan \alpha$?
 (a) $1/2$ (b) $1/3$
 (c) $1/4$ (d) None of these
- 31.** If $\alpha = 30^\circ$, then what is the value of $\tan \beta$?
 (a) 1 (b) $1/2$
 (c) $1/3$ (d) None of these
- 32.** If $\alpha = 30^\circ$ and $h = 30$ m, then what is the distance between the base of the building and the base of the tower?
 (a) $(15 + 15\sqrt{3})$ m (b) $(30 + 15\sqrt{3})$ m
 (c) $(45 + 15\sqrt{3})$ m (d) None of these
- 33.** If $\beta = 30^\circ$ and if θ is the angle of depression of the foot of the tower as seen from the top of the building, then what is the value of $\tan \theta$?
 (a) $\frac{(3 - \sqrt{3})}{3\sqrt{3}}$ (b) $\frac{(3 + \sqrt{3})}{3\sqrt{3}}$
 (c) $\frac{(2 - \sqrt{3})}{3\sqrt{3}}$ (d) None of these

- 34.** What is the angle of elevation of the Sun, when the shadow of a pole of height x m is $\frac{x}{\sqrt{3}}$ m? ☑ 2013 II
 (a) 30° (b) 45° (c) 60° (d) 75°
- 35.** A spherical balloon of radius r subtends angle 60° at the eye of an observer. If the angle of elevation of its centre is 60° and h is the height of the centre of the balloon, then which one of the following is correct? ☑ 2013 II
 (a) $h = r$ (b) $h = \sqrt{2}r$
 (c) $h = \sqrt{3}r$ (d) $h = 2r$
- 36.** The angle of elevation of the top of a tower 30 m high from the foot of another tower in the same plane is 60° and the angle of elevation of the top of the second tower from the foot of the first tower is 30° . The distance between the two towers is m times the height of the shorter tower. What is m equal to? ☑ 2014 I
 (a) $\sqrt{2}$ (b) $\sqrt{3}$ (c) $\frac{1}{2}$ (d) $\frac{1}{3}$
- 37.** The shadow of a tower standing on a level plane is found to be 50 m longer when the Sun's elevation is 30° , then when it is 60° . What is the height of the tower? ☑ 2014 I
 (a) 25 m (b) $25\sqrt{3}$ m (c) $\frac{25}{\sqrt{3}}$ m (d) 30 m
- 38.** From a certain point on a straight road, a person observe a tower in the West direction at a distance of 200 m. He walks some distance along the road and finds that the same tower is 300 m South of him. What is the shortest distance of the tower from the road? ☑ 2014 II
 (a) $\frac{300}{\sqrt{13}}$ m (b) $\frac{500}{\sqrt{13}}$ m (c) $\frac{600}{\sqrt{13}}$ m (d) $\frac{900}{\sqrt{13}}$ m
- 39.** If from the top of a post a string twice the length of the post is stretched tight to a point on the ground, then what angle will the string make with the post? ☑ 2014 II
 (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{5\pi}{12}$ (d) $\frac{\pi}{3}$
- 40.** The angle of elevation of a cloud from a point 200 m above a lake is 30° and the angle of depression of its reflection in the lake is 60° . The height of the cloud is ☑ 2015 I
 (a) 200 m (b) 300 m (c) 400 m (d) 600 m
- 41.** From the top of a tower, the angles of depression of two objects P and Q (situated on the ground on the same side of the tower) separated at a distance of $100(3 - \sqrt{3})$ m are 45° and 60° , respectively. The height of the tower is ☑ 2015 I
 (a) 200 m (b) 250 m
 (c) 300 m (d) None of these

- 42.** The angles of elevation of the top of a tower from two points P and Q at distance m^2 and n^2 respectively, from the base and in the same straight line with it are complementary. The height of the tower is **2015 I**

(a) $(mn)^{1/2}$ (b) $mn^{1/2}$
(c) $m^{1/2}n$ (d) mn

- 43.** A pole is standing erect on the ground which is horizontal. The tip of the pole is tied tight with a rope of length $\sqrt{12}$ m to a point on the ground. If the rope is making 30° with the horizontal, then the height of the pole is **2015 II**

(a) $2\sqrt{3}$ m (b) $3\sqrt{2}$ m
(c) 3 m (d) $\sqrt{3}$ m

- 44.** An aeroplane flying at a height of 3000 m passes vertically above another aeroplane at an instant, when the angles of elevation of the two planes from some point on the ground are 60° and 45° , respectively. Then, the vertical distance between the two planes is **2015 II**

(a) $1000(\sqrt{3}-1)$ m (b) $1000\sqrt{3}$ m
(c) $1000(3-\sqrt{3})$ m (d) $3000\sqrt{3}$ m

- 45.** Two observers are stationed due North of a tower (of height x m) at a distance y m from each other. The angles of elevation of the tower observed by them are 30° and 45° , respectively. Then, x/y is equal to **2016 I**

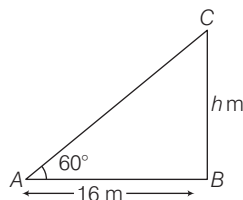
(a) $\frac{\sqrt{2}-1}{2}$ (b) $\frac{\sqrt{3}-1}{2}$ (c) $\frac{\sqrt{3}+1}{2}$ (d) 1

ANSWERS

1	c	2	a	3	c	4	c	5	c	6	b	7	b	8	d	9	d	10	b
11	c	12	b	13	d	14	a	15	a	16	b	17	c	18	c	19	c	20	b
21	c	22	a	23	a	24	a	25	b	26	a	27	d	28	c	29	c	30	b
31	a	32	c	33	a	34	c	35	c	36	b	37	b	38	c	39	d	40	c
41	c	42	d	43	d	44	c	45	c										

HINTS AND SOLUTIONS

- 1.** (c) Let the height of the tree be h m.
 $\therefore BC = h$ m and $AB = 16$ m



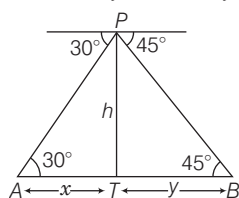
In right angled $\triangle ABC$,

$$\tan 60^\circ = \frac{BC}{AB} = \frac{h}{16} \Rightarrow \sqrt{3} = \frac{h}{16}$$

$$\Rightarrow h = 16\sqrt{3} \text{ m}$$

Hence, the height of the tree is $16\sqrt{3}$ m.

- 2.** (a) In right angled $\triangle PBT$, $\tan 45^\circ = \frac{h}{y} \Rightarrow 1 = \frac{h}{y}$



$$\therefore y = h \quad \dots(i)$$

and in right angled $\triangle PTA$,

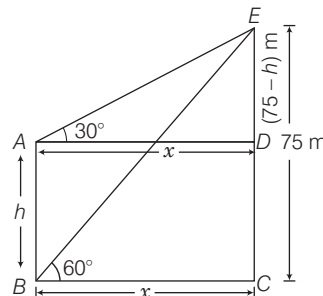
$$\tan 30^\circ = \frac{h}{x} \Rightarrow x = \sqrt{3}h \quad \dots(ii)$$

$\therefore$ Required distance, $AB = x + y$

$$x + y = \sqrt{3}h + h = h(\sqrt{3} + 1) \text{ m}$$

- 3.** (c) Let height of the building be h m and distance between building and tower be x m.

$\therefore AB = h$ m and $BC = x$ m



In right angled $\triangle ADE$,

$$\tan 30^\circ = \frac{ED}{AD} \Rightarrow \frac{1}{\sqrt{3}} = \frac{75-h}{x}$$

$$\Rightarrow x = 75\sqrt{3} - h\sqrt{3} \quad \dots(i)$$

and in right angled $\triangle BCE$,

$$\tan 60^\circ = \frac{CE}{BC} \Rightarrow \sqrt{3} = \frac{75}{x}$$

$$\Rightarrow x\sqrt{3} = 75$$

$$\Rightarrow (75\sqrt{3} - h\sqrt{3})\sqrt{3} = 75 \text{ [from Eq. (i)]}$$

$$\Rightarrow 75 \times 3 - 3h = 75$$

$$\Rightarrow 3h = 75 \times 3 - 75 \Rightarrow h = \frac{75 \times 2}{3}$$

$$\therefore h = 50 \text{ m}$$

Hence, the height of the building is 50 m.

- 4.** (c) Let BC be a building of height h m and CD be a height of antenna. and let distance between A and $B = x$ m

